| EXPERIMENT | IMPLEMENTATION OF MAP REDUCE PARADIGM | ROLL NO | 23DI16 | |
|---|---|---|---|---|
| | | DAY | MONTH | YEAR |
| 3 | | | | |

**AIM:**

To implement the map reduce paradigm.

**COMPONENTS REQUIRED:**

| S.no | FACILITIES REQUIRED | QUANTITY |
|---|---|---|
| 1 | System | 01 |
| 2 | Operating System | Ubuntu |

**PROCEDURE:**

**Step 1:** Start the Hadoop Services

**Step 2:** Prepare the Input Data for Processing

**Step 3:** Upload the Input File to HDFS

**Step 4:** Execute the MapReduce WordCount Job.

**Step 5:** View and Validate Output from HDFS

**Step 6:** Analyze Results and Complete Cleanup.

**OUTPUT:**

```
dit@dit-OptiPlex-3060:~$ sudo apt update
[sudo] password for dit:
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [752 k
B]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [934
kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,90
3 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,1
64 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [418
kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metada
ta [54.6 kB]
```

```
dit@dit-OptiPlex-3060:~$ su - hdoop
Password:
hdoop@dit-OptiPlex-3060:~$ cd ~
mvn archetype:generate \
-DgroupId=com.hdoop.wordcount \
-DartifactId=wordcount \
-DarchetypeArtifactId=maven-archetype-quickstart \
-DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-source
s @ standalone-pom >>>
```

```
dit@dit-OptiPlex-3060:~$ sudo apt install maven -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
maven is already the newest version (3.6.3-5).
0 upgraded, 0 newly installed, 0 to remove and 241 not upgraded.
```

```
hdoop@dit-OptiPlex-3060:~$ cd wordcount
hdoop@dit-OptiPlex-3060:~/wordcount$ nano pom.xml
hdoop@dit-OptiPlex-3060:~/wordcount$ nano src/main/java/com/hdoop/wordcount/Word
Count.java
hdoop@dit-OptiPlex-3060:~/wordcount$ rm -rf src/test
hdoop@dit-OptiPlex-3060:~/wordcount$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.hdoop.wordcount:wordcount >--------------------
[INFO] Building wordcount 1.0
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ wordcount
---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i
.e. build is platform dependent!
```

```
hdoop@dit-OptiPlex-3060:~/wordcount$ nano data.txt
```

```
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -ls /user/
Found 1 items
drwxr-xr-x   - hdoop supergroup          0 2026-01-20 20:32 /user/hdoop
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -put data.txt /user/hdoop/input/
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -ls /user/hdoop/input/
Found 1 items
-rw-r--r--   1 hdoop supergroup         45 2026-01-20 20:37 /user/hdoop/input/data.txt
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -mkdir -p /user/hadoop/input
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -put data.txt /user/hadoop/input/
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -ls /user/hadoop/input/
Found 1 items
-rw-r--r--   1 hdoop supergroup         45 2026-01-20 20:38 /user/hadoop/input/data.txt
```

```
hdoop@dit-OptiPlex-3060:~/wordcount$ hadoop jar target/wordcount-1.0.jar \
/user/hdoop/wordcount/input \
/user/hdoop/wordcount/output
2026-01-20 20:40:15,290 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /127.0.0.1:8032
2026-01-20 20:40:15,731 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the
with ToolRunner to remedy this.
2026-01-20 20:40:15,754 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hdo
2026-01-20 20:40:16,021 INFO input.FileInputFormat: Total input files to process : 3
2026-01-20 20:40:16,189 INFO mapreduce.JobSubmitter: number of splits:3
2026-01-20 20:40:16,972 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1768917475497_0001
2026-01-20 20:40:16,972 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-20 20:40:17,118 INFO conf.Configuration: resource-types.xml not found
2026-01-20 20:40:17,118 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-20 20:40:17,508 INFO impl.YarnClientImpl: Submitted application application_1768917475497_0001
2026-01-20 20:40:17,577 INFO mapreduce.Job: The url to track the job: http://dit-OptiPlex-3060:8088/proxy/application_1768
2026-01-20 20:40:17,577 INFO mapreduce.Job: Running job: job_1768917475497_0001
2026-01-20 20:40:22,683 INFO mapreduce.Job: Job job_1768917475497_0001 running in uber mode : false
2026-01-20 20:40:22,685 INFO mapreduce.Job:  map 0% reduce 0%
2026-01-20 20:40:26,826 INFO mapreduce.Job:  map 100% reduce 0%
2026-01-20 20:40:31,863 INFO mapreduce.Job:  map 100% reduce 100%
2026-01-20 20:40:32,877 INFO mapreduce.Job: Job job_1768917475497_0001 completed successfully
2026-01-20 20:40:32,938 INFO mapreduce.Job: Counters: 54
        File System Counters
```

```
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -cat /user/hdoop/wordcount/output/part-r-00000
3ex     1
ds      1
dwarakesh,shyam,aagash  1
for     1
output  1
hdoop@dit-OptiPlex-3060:~/wordcount$
```

**RESULT :**

Thus, the implementation of map reduce paradigm has been implemented successfully.

```
dit@dit-OptiPlex-3060:~$ sudo apt update
[sudo] password for dit:
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [752 k
B]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [934
kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,90
3 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,1
64 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [418
kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metada
ta [54.6 kB]
```

```
dit@dit-OptiPlex-3060:~$ su - hdoop
Password:
hdoop@dit-OptiPlex-3060:~$ cd ~
mvn archetype:generate \
-DgroupId=com.hdoop.wordcount \
-DartifactId=wordcount \
-DarchetypeArtifactId=maven-archetype-quickstart \
-DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-source
s @ standalone-pom >>>
```

```
dit@dit-OptiPlex-3060:~$ sudo apt install maven -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
maven is already the newest version (3.6.3-5).
0 upgraded, 0 newly installed, 0 to remove and 241 not upgraded.
```

```
hdoop@dit-OptiPlex-3060:~$ cd wordcount
hdoop@dit-OptiPlex-3060:~/wordcount$ nano pom.xml
hdoop@dit-OptiPlex-3060:~/wordcount$ nano src/main/java/com/hdoop/wordcount/Word
Count.java
hdoop@dit-OptiPlex-3060:~/wordcount$ rm -rf src/test
hdoop@dit-OptiPlex-3060:~/wordcount$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.hdoop.wordcount:wordcount >-------------------
[INFO] Building wordcount 1.0
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ wordcount
---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i
.e. build is platform dependent!
```

```
hdoop@dit-OptiPlex-3060:~/wordcount$ nano data.txt
```

```
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -ls /user/
Found 1 items
drwxr-xr-x   - hdoop supergroup          0 2026-01-20 20:32 /user/hdoop
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -put data.txt /user/hdoop/input/
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -ls /user/hdoop/input/
Found 1 items
-rw-r--r--   1 hdoop supergroup         45 2026-01-20 20:37 /user/hdoop/input/data.txt
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -mkdir -p /user/hadoop/input
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -put data.txt /user/hadoop/input/
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -ls /user/hadoop/input/
Found 1 items
-rw-r--r--   1 hdoop supergroup         45 2026-01-20 20:38 /user/hadoop/input/data.txt
```

```
hdoop@dit-OptiPlex-3060:~/wordcount$ hadoop jar target/wordcount-1.0.jar \
/user/hdoop/wordcount/input \
/user/hdoop/wordcount/output
2026-01-20 20:40:15,290 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /127.0.0.1:8032
2026-01-20 20:40:15,731 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the
with ToolRunner to remedy this.
2026-01-20 20:40:15,754 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hdo
2026-01-20 20:40:16,021 INFO input.FileInputFormat: Total input files to process : 3
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2026-01-20 20:40:16,972 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-01-20 20:40:17,118 INFO conf.Configuration: resource-types.xml not found
2026-01-20 20:40:17,118 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-01-20 20:40:17,508 INFO impl.YarnClientImpl: Submitted application application_1768917475497_0001
2026-01-20 20:40:17,577 INFO mapreduce.Job: The url to track the job: http://dit-OptiPlex-3060:8088/proxy/application_1768
2026-01-20 20:40:17,577 INFO mapreduce.Job: Running job: job_1768917475497_0001
2026-01-20 20:40:22,683 INFO mapreduce.Job: Job job_1768917475497_0001 running in uber mode : false
2026-01-20 20:40:22,685 INFO mapreduce.Job:  map 0% reduce 0%
2026-01-20 20:40:26,826 INFO mapreduce.Job:  map 100% reduce 0%
2026-01-20 20:40:31,863 INFO mapreduce.Job:  map 100% reduce 100%
2026-01-20 20:40:32,877 INFO mapreduce.Job: Job job_1768917475497_0001 completed successfully
2026-01-20 20:40:32,938 INFO mapreduce.Job: Counters: 54
        File System Counters
```

```
hdoop@dit-OptiPlex-3060:~/wordcount$ hdfs dfs -cat /user/hdoop/wordcount/output/part-r-00000
3ex     1
ds      1
dwarakesh,shyam,aagash  1
for     1
output  1
hdoop@dit-OptiPlex-3060:~/wordcount$
```

| EXPERIMENT | INSTALL AND RUN PIG | ROLL NO | 23DI16 | |
|---|---|---|---|---|
| | | DAY | MONTH | YEAR |
| 4 | | | | |

**AIM:**

To install Apache Pig on Hadoop in Ubuntu by downloading the Pig package, extracting it, configuring environment variables, and launching the Pig Grunt shell.

**COMPONENTS REQUIRED:**

| S.no | FACILITIES REQUIRED | QUANTITY |
|---|---|---|
| 1 | System | 01 |
| 2 | Operating System | Ubuntu |

**PROCEDURE:**

**Step 1:** Download Apache Pig using "wget".

***$~ wget https://downloads.apache.org/pig/pig-0.17.0/pig-0.17.0.tar.gz***

```
dit@dit-OptiPlex-3060:~$ su - hdoop
Password:
hdoop@dit-OptiPlex-3060:~$ cd
hdoop@dit-OptiPlex-3060:~$ ls
Desktop              names1.txt           pig-0.17.0             Public
dfsdata              names.txt            pig-0.17.0.tar.gz      snap
Documents            Pictures             pig-0.17.0.tar.gz.1    Templates
Downloads            pig                  pig_1767103639756.log  tmpdata
filter_names.pig     pig-0.15.0.tar.gz    pig_1767105789409.log  Videos
hadoop-3.4.0         pig-0.16.0.tar.gz    pig_1767108576703.log  wget-log
hadoop-3.4.0.tar.gz  pig-0.16.0.tar.gz.1  pig_1767108934723.log  wordcount
hadoopdata           pig-0.16.0.tar.gz.2  pig_1767108940059.log  WordCount
Music                pig-0.16.0.tar.gz.3  pig_1767109475060.log  WordCountLab
hdoop@dit-OptiPlex-3060:~$  wget https://downloads.apache.org/pig/pig-0.17.0/pig
-0.17.0.tar.gz
--2026-01-20 21:24:20--  https://downloads.apache.org/pig/pig-0.17.0/pig-0.17.0.
tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.
214.104, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 230606579 (220M) [application/x-gzip]
Saving to: 'pig-0.17.0.tar.gz.2'

pig-0.17.0.tar.gz.2   5%[>                    ]  12.94M  1.38MB/s    eta 2m 40s
```

**Step 2 :** Extract Pig

Extract the downloaded Pig tar file.

***$~ tar -xvzf pig-0.17.0.tar.gz***

```
hdoop@dit-OptiPlex-3060:~$ tar -xvzf pig-0.17.0.tar.gz
```

```
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/physicalLayer/expre
ssionOperators/regex/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/physicalLayer/plans
/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/physicalLayer/relat
ionalOperators/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/physicalLayer/util/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/converter/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/operator/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/optimizer/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/plan/
```

**Step 3 :** Move Pig Directory

***$~ mv pig-0.17.0 pig ls***

```
hdoop@dit-OptiPlex-3060:~$ mv pig-0.17.0 pig
```

**Step 4 :** Set Pig Environment Variables

Open ".bashrc" file and add Pig environment variables.

***$~ nano ~/.bashrc***

```
hdoop@dit-OptiPlex-3060:~$ nano .bashrc
hdoop@dit-OptiPlex-3060:~$
```

**(Add below lines)** export PIG_HOME=/home/hdoop/pig export PATH=$PATH:$PIG_HOME/bin

```
GNU nano 6.2                    /home/hdoop/.bashrc

# Hadoop
export HADOOP_HOME=/home/hdoop/hadoop-3.4.0
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

# Pig
export PIG_HOME=/home/hdoop/pig-0.17.0
export PATH=$PATH:$PIG_HOME/bin

# PIG ENV
export PIG_HOME=/home/hdoop/pig
export PATH=$PATH:$PIG_HOME/bin

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

**Step** 5 : Apply the Environment changes.
Reload the .bashrc file to apply changes.

***$~* source ~/.bashrc**

```
hdoop@dit-OptiPlex-3060:~$ source ~/.bashrc
```

**Step 6 :** Check pig version.

***$~ pig -version***

```
hdoop@dit-OptiPlex-3060:~$ pig -version
Apache Pig version 0.16.0 (r1746530)
compiled Jun 01 2016, 23:10:49
```

**Step 7 :** Start pig shell . Pig

***$~ pig***

```
hdoop@dit-OptiPlex-3060:~$ pig
2026-01-20 21:21:59,971 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-01-20 21:21:59,971 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-01-20 21:21:59,971 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecT
ype
2026-01-20 21:21:59,992 [main] INFO  org.apache.pig.Main - Apache Pig version 0.
16.0 (r1746530) compiled Jun 01 2016, 23:10:49
2026-01-20 21:21:59,992 [main] INFO  org.apache.pig.Main - Logging error message
s to: /home/hdoop/pig_1768924319989.log
2026-01-20 21:22:00,018 [main] INFO  org.apache.pig.impl.util.Utils - Default bo
otup file /home/hdoop/.pigbootup not found
2026-01-20 21:22:00,185 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
ess
2026-01-20 21:22:00,185 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-01-20 21:22:00,185 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.HExecutionEngine - Connecting to hadoop file system at: hdfs://127.0.0.1:9000
2026-01-20 21:22:00,496 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-01-20 21:22:00,507 [main] INFO  org.apache.pig.PigServer - Pig Script ID fo
r the session: PIG-default-6093cffc-8c1e-4912-aa60-12c5c89e92aa
2026-01-20 21:22:00,507 [main] WARN  org.apache.pig.PigServer - ATS is disabled
since yarn.timeline-service.enabled set to false
grunt>
```

**RESULT:**

Thus, to install Apache Pig on Hadoop in Ubuntu by downloading the Pig package, extracting it, configuring environment variables, and launching the Pig Grunt shell has been implemented successfully.

```
dit@dit-OptiPlex-3060:~$ su - hdoop
Password:
hdoop@dit-OptiPlex-3060:~$ cd
hdoop@dit-OptiPlex-3060:~$ ls
Desktop                 names1.txt            pig-0.17.0                Public
dfsdata                 names.txt             pig-0.17.0.tar.gz         snap
Documents               Pictures              pig-0.17.0.tar.gz.1       Templates
Downloads               pig                   pig_1767103639756.log     tmpdata
filter_names.pig        pig-0.15.0.tar.gz     pig_1767105789409.log     Videos
hadoop-3.4.0            pig-0.16.0.tar.gz     pig_1767108576703.log     wget-log
hadoop-3.4.0.tar.gz     pig-0.16.0.tar.gz.1   pig_1767108934723.log     wordcount
hadoopdata              pig-0.16.0.tar.gz.2   pig_1767108940059.log     WordCount
Music                   pig-0.16.0.tar.gz.3   pig_1767109475060.log     WordCountLab
hdoop@dit-OptiPlex-3060:~$  wget https://downloads.apache.org/pig/pig-0.17.0/pig
-0.17.0.tar.gz
--2026-01-20 21:24:20--  https://downloads.apache.org/pig/pig-0.17.0/pig-0.17.0.
tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.
214.104, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 230606579 (220M) [application/x-gzip]
Saving to: ‘pig-0.17.0.tar.gz.2’

pig-0.17.0.tar.gz.2   5%[>                    ]  12.94M  1.38MB/s    eta 2m 40s
```

```
hdoop@dit-OptiPlex-3060:~$ tar -xvzf pig-0.17.0.tar.gz
```

```
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/physicalLayer/expre
ssionOperators/regex/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/physicalLayer/plans
/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/physicalLayer/relat
ionalOperators/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/physicalLayer/util/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/converter/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/operator/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/optimizer/
pig-0.17.0/src/org/apache/pig/backend/hadoop/executionengine/spark/plan/
```

```
hdoop@dit-OptiPlex-3060:~$ mv pig-0.17.0 pig
```

```
hdoop@dit-OptiPlex-3060:~$ nano .bashrc
hdoop@dit-OptiPlex-3060:~$
```

```
  GNU nano 6.2                    /home/hdoop/.bashrc

# Hadoop
export HADOOP_HOME=/home/hdoop/hadoop-3.4.0
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

# Pig
export PIG_HOME=/home/hdoop/pig-0.17.0
export PATH=$PATH:$PIG_HOME/bin

# PIG ENV
export PIG_HOME=/home/hdoop/pig
export PATH=$PATH:$PIG_HOME/bin

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
hdoop@dit-OptiPlex-3060:~$ source ~/.bashrc
```

```
hdoop@dit-OptiPlex-3060:~$ pig -version
Apache Pig version 0.16.0 (r1746530)
compiled Jun 01 2016, 23:10:49
```

```
hdoop@dit-OptiPlex-3060:~$ pig
2026-01-20 21:21:59,971 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-01-20 21:21:59,971 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-01-20 21:21:59,971 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecT
ype
2026-01-20 21:21:59,992 [main] INFO  org.apache.pig.Main - Apache Pig version 0.
16.0 (r1746530) compiled Jun 01 2016, 23:10:49
2026-01-20 21:21:59,992 [main] INFO  org.apache.pig.Main - Logging error message
s to: /home/hdoop/pig_1768924319989.log
2026-01-20 21:22:00,018 [main] INFO  org.apache.pig.impl.util.Utils - Default bo
otup file /home/hdoop/.pigbootup not found
2026-01-20 21:22:00,185 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
ess
2026-01-20 21:22:00,185 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-01-20 21:22:00,185 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.HExecutionEngine - Connecting to hadoop file system at: hdfs://127.0.0.1:9000
2026-01-20 21:22:00,496 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
2026-01-20 21:22:00,507 [main] INFO  org.apache.pig.PigServer - Pig Script ID fo
r the session: PIG-default-6093cffc-8c1e-4912-aa60-12c5c89e92aa
2026-01-20 21:22:00,507 [main] WARN  org.apache.pig.PigServer - ATS is disabled
since yarn.timeline-service.enabled set to false
grunt>
```